

DATA MINING

AND

WRANGLING

ASSIGNMENT

2 / 9 / 2026

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DATA MINING AND WRANGLING ASSIGNMENT

1. Question One.

- a) Support is a measure of how frequent an item appears in the entire dataset. It tells us how common a product or a combination of products occurs.

$$\text{Support}(B) = \frac{\text{number of transactions containing } B}{\text{Total no of transactions}}$$

b) Confidence

Confidence measures how likely an event will occur when another event has already happened.

$$\text{Confidence}(B \rightarrow A) = \frac{\text{Support}_{A+B}}{\text{Support}_B}$$

c) Lift

Lift measures whether two items occur together more often than expected by chance. It shows the usefulness of an association rule.

$$\text{Lift}(B \rightarrow A) = \frac{\text{Confidence}(B \rightarrow A)}{\text{Support}_A}$$

Lift value Interpretation:

Lift = 1 - Items occur together exactly as expected.

Lift > 1 - positive association: Items occur together more often than expected.

Lift < 1 - negative association: Items occur together less often than expected.

Question 2

No, The rule is not considered interesting

-Confidence = 0.85 (85)% means that when the antecedent occurs, the consequent occurs 85% of the time.

-Lift = 0.7 means the consequent is actually less likely to occur with the antecedent than expected by random chance.

Although the rule looks strong because of its high confidence, the lift shows the association is negative.

A lift below 1 indicates the rule does not reveal a meaningful relationship between the items.

Question 3.

The Apriori principle states that if an itemset is frequent, then ~~the~~ all its subsets must also be frequent.

Question 4

Apriori algorithm is a data mining algorithm used to discover frequent itemsets and association rules for transaction data. Its purpose is to identify combinations of items that frequently occur together in a dataset.

Question 5.

are

a) C represents candidate itemsets. These possible item combinations generated during each apriori iteration. They have not yet been confirmed as frequent.

b) L represents frequent itemsets. These are candidate itemsets whose support count meets or exceeds the minimum support ~~threshold~~ threshold. These survive pruning and are used to generate large candidates.

Question 6

Join step is where Apriori combines smaller frequent itemsets to create larger possible itemsets

Prune step checks whether a candidate item is worth testing. If any smaller subset of that candidate is not frequent, the candidate is immediately removed. This saves time and computation.

Question 7

Item	Support Count	Support fraction
Bread	6	0.75 (75%)
Milk	6	0.75 (75%)
Butter	5	0.625 (62.5%)
Eggs	4	0.50 (50%)

$$\text{Support of Bread} = \frac{6}{8} = 0.75$$

$$\text{Support of Milk} = \frac{6}{8} = 0.75$$

$$\text{Support of Butter} = \frac{5}{8} = 0.625$$

$$\text{Support of Eggs} = \frac{4}{8} = 0.50$$

Question 8.

L1

L1 Itemset	Support Count
Bread	6
Milk	6
Butter	5
Eggs	4

C2.

Candidate Pair	Support Count	Frequent.
{Bread, Milk}	4	Yes
{Bread, Butter}	4	Yes
{Bread, Eggs}	3	Yes
{Milk, Butter}	4	Yes
{Milk, Eggs}	2	No
{Butter, Eggs}	2	No

L2

L2 Frequent Pair	Support Count
{Bread, Milk}	4
{Bread, Butter}	4
{Bread, eggs}	3
{Milk, Butter}	4

C3

Candidate Pair	Support Count	Frequent
{Bread, Milk, Butter}	3	Yes
{Bread, Butter, Eggs}	2	No
{Butter, Milk, Eggs}	1	No

L3

L3 Frequent Pair	Support count
{Bread, Milk, Butter}	3

Question 7.

Itemset

Support count

{Bread, Milk, Butter}

{Bread, Milk}

Question 9.

Candidate removed

{Bread, Butter, Eggs}

Why is it removed?

Subsets	Support Count	Frequent.
{Bread, Milk}	4	Yes
{Bread, Eggs}	2	No
{Milk, Eggs}	2	No

Infrequent subset responsible

{Bread, Eggs} or {Milk, Eggs}, since both are infrequent.

Apriori says every subset must be frequent, since Bread and Eggs is not frequent, Bread, Milk, Eggs is pruned before counting.

Question 10

Itemset	Support Count
{Bread, Milk} $\rightarrow$ {Butter}	

Part	Answer
Antecedent	{Bread, Milk}
Consequent	{Butter}

The antecedent is the "if" part of the rule, while the consequent is the "then" part.

ii) Itemset	Support count.
{Bread, Milk, Butter}	3
{Bread, Milk}	4
{Bread, Butter}	4
{Milk, Butter}	4
{Bread}	6
{Milk}	6
{Butter}	5

Confidence calculations:

Rule	Calculation	Confidence
$\{ \text{Bread, Milk} \} \rightarrow \text{Butter}$	$\frac{3}{4}$	0.75
$\{ \text{Bread, Butter} \} \rightarrow \text{Milk}$	$\frac{3}{4}$	0.75
$\{ \text{Milk, Butter} \} \rightarrow \text{Bread}$	$\frac{3}{4}$	0.75
$\{ \text{Bread} \} \rightarrow \{ \text{Milk, Butter} \}$	$\frac{3}{6}$	0.50
$\{ \text{Milk} \} \rightarrow \{ \text{Bread, Butter} \}$	$\frac{3}{6}$	0.50
$\{ \text{Butter} \} \rightarrow \{ \text{Bread, Milk} \}$	$\frac{3}{5}$	0.60

ii) Rule	Confidence
$\{ \text{Bread, Milk} \} \rightarrow \text{Butter}$	75%
$\{ \text{Bread, Butter} \} \rightarrow \text{Milk}$	75%
$\{ \text{Milk, Butter} \} \rightarrow \text{Bread}$	75%
$\{ \text{Bread} \} \rightarrow \{ \text{Milk, Butter} \}$	50%
$\{ \text{Milk} \} \rightarrow \{ \text{Bread, Butter} \}$	50%
$\{ \text{Butter} \} \rightarrow \{ \text{Bread, Milk} \}$	60%

A rule is strong if:

$$\text{Confidence} \geq \text{Minimum Confidence}$$